

EXPERIMENT NO. 3

Date: _____

TITLE

Pandas DataFrame – Student Details

AIM

To create a student dataframe, display rows/columns, filter rows, compute summary statistics with department-wise average, and save it as CSV.

OBJECTIVE

To work with a Pandas DataFrame of five students (roll no, name, department, marks) and perform selection, filtering, aggregation and CSV export operations.

THEORY

DataFrame with rows and named columns:

`df = pd.DataFrame({columns})` `df.head(n)`, `df.tail(n)`

Column selection and boolean filtering:

`df[["Name", "Marks"]]` `df[df["Marks"] > 80]`

Summary statistics:

$$\text{max, min, mean} = \frac{1}{n} \sum x_i, \text{ sum} = \sum x_i$$

Department-wise average and CSV export:

`df.groupby("Department")["Marks"].mean()` `df.to_csv(file)`

PROCEDURE

1. Create a DataFrame of 5 students with Roll no, Name, Department, Marks and display it.
2. Display the first 3 rows (head), last 2 rows (tail) and only Name/Marks columns.
3. Filter students with marks above 80 and students of CSE department.
4. Compute highest, lowest, average and total marks; average mark department-wise.
5. Save the DataFrame to CSV and verify the file.

DATASET DESCRIPTION

Dataset Name: Student details (entered in program)

Source: 5 students; departments CSE, ECE, ME

Number of Samples: 5 rows Number of Features: 4 columns

Target / Class: Not applicable (data handling operations)

Description: Marks range 68–90; exported to exp3_students.csv.

PROGRAM

```
import pandas as pd

data = {
    "Roll no": [1, 2, 3, 4, 5],
    "Name": ["Anu", "Ravi", "Meera", "Arjun", "Divya"],
    "Department": ["CSE", "ECE", "CSE", "ME", "CSE"],
    "Marks": [85, 72, 90, 68, 78],
}

df = pd.DataFrame(data)
print("Dataframe:")
print(df.to_string(index=False))

print("\nFirst 3 rows:")
print(df.head(3).to_string(index=False))
print("\nLast two rows:")
print(df.tail(2).to_string(index=False))
print("\nName and Marks columns:")
print(df[["Name", "Marks"]].to_string(index=False))
print("\nMarks more than 80:")
print(df[df["Marks"] > 80].to_string(index=False))
print("\nCSE department:")
print(df[df["Department"] == "CSE"].to_string(index=False))
print("\nHighest =", df["Marks"].max(), " Lowest =", df["Marks"].min(),
      " Avg =", round(df["Marks"].mean(), 2), " Total =", df["Marks"].sum())
print("\nAverage mark department wise:")
print(df.groupby("Department")["Marks"].mean().round(2).to_string())

df.to_csv("exp3_students.csv", index=False)
print("\nSaved to exp3_students.csv")
```

OUTPUT

Dataframe:

Roll no	Name	Department	Marks
1	Anu	CSE	85
2	Ravi	ECE	72
3	Meera	CSE	90
4	Arjun	ME	68
5	Divya	CSE	78

First 3 rows:

Roll no	Name	Department	Marks
1	Anu	CSE	85
2	Ravi	ECE	72
3	Meera	CSE	90

Last two rows:

Roll no	Name	Department	Marks
4	Arjun	ME	68
5	Divya	CSE	78

Name and Marks columns:

Name	Marks
Anu	85
Ravi	72
Meera	90
Arjun	68
Divya	78

Marks more than 80:

Roll no	Name	Department	Marks
1	Anu	CSE	85
3	Meera	CSE	90

CSE department:

Roll no	Name	Department	Marks
1	Anu	CSE	85
3	Meera	CSE	90
5	Divya	CSE	78

Highest = 90 Lowest = 68 Avg = 78.6 Total = 393

Average mark department wise:

Department	Average Mark
CSE	84.33
ECE	72.00
ME	68.00

Saved to exp3_students.csv

RESULT

The above program was implemented and executed successfully using Python, and the corresponding output was obtained and verified. The results of the **Pandas DataFrame operations** were analyzed and found to be satisfactory, thereby achieving the desired objective of the experiment.

Faculty Signature: _____

Date: _____